

Webinar – 1: Problem Sheet

Problem 1

You are given a system of three linear equations:

$$\begin{aligned}\alpha x + 2y + 6z &= b_1, \\ \alpha x + \alpha y + 5z &= b_2, \\ \alpha x + \alpha y + \alpha z &= b_3.\end{aligned}$$

In each equation, α is an unknown constant.

Task: Find the three possible values of α for which the system fails to have three pivots. Give a clear reason in each case.

Problem 2

Let

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$

- Using elementary row operations, reduce $\mathbf{A}$ to its row echelon form (REF).
- Let $\mathbf{V}$ be the column space of $\mathbf{A}$. Find a basis for $\mathbf{V}$ and state its dimension.
- Let $\mathbf{V}$ be the null space of $\mathbf{A}$, i.e., the set of all $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{0}$. Find a basis for this space and its dimension.
- Find a maximal linearly independent set of rows of $\mathbf{A}$ and state how many such rows there are.

Problem 3

Let $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$, where $\mathbf{C}_m \in \mathbb{R}^4$ are the columns of $\mathbf{A}$. Suppose:

- $\text{rank}(\mathbf{A}) = 2$,
- $\mathbf{C}_2 = 3\mathbf{C}_1$,

- $\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3$,
- A particular solution of $\mathbf{Ax} = \mathbf{b}$ is $[1, 0, 1, 0]^T$.

(a) Find the general solution of $\mathbf{Ax} = \mathbf{b}$.

(b) If $\text{RREF}(\mathbf{A}) = \mathbf{A}$, determine $\mathbf{b}$.

Problem 4

Let

$$M = \{\mathbf{A} \in \mathbb{R}^{2 \times 2} \mid \mathbf{A} = -\mathbf{A}^T\}.$$

(a) Prove that M is a subspace of the vector space of all 2×2 real matrices.

(b) Decide whether the following set is a basis for M :

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

(c) Determine whether the following matrices are linearly independent:

$$\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} \right\}.$$

Problem 5

Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^D$ be linearly independent with $D > 3$.

You want to construct a $D \times 3$ matrix whose columns are linear combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Task:

Either:

- Construct such a matrix with full column rank, *or*
- Give a mathematical argument explaining why this is impossible.

Problem 6

Let

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & -2 \end{bmatrix},$$

and define

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T \mathbf{A} \mathbf{y}, \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^4.$$

(a) Prove that this defines an inner product on $\mathbb{R}^4$.

(b) Find all vectors $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ such that

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$$

where $\mathbf{x} = [1, 0, 0, 0]^T$

Problem 7

Using the Cauchy–Schwarz inequality, prove that

$$(a_1 + a_2 + \cdots + a_n)^2 \leq n(a_1^2 + a_2^2 + \cdots + a_n^2)$$

for arbitrary real numbers $a_1, a_2, \dots, a_n$.

Also determine when equality holds.

Problem 8

Consider the positive definite matrix

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}.$$

(i) Find a lower triangular matrix L such that $A = LL^T$. Then compute all three eigenvalues of L .

(ii) Let $\mathbf{z} = L^T \mathbf{x}$ where $x \neq 0$. Prove that $\mathbf{z}$ can never be the zero vector.

Problem 9

(a) Matrix with zero row sums

Let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ satisfy

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for all } i = 1, \dots, n.$$

- (i) Can 0 be an eigenvalue of A ? Justify your answer.
- (ii) Compute $\det(A)$.
- (iii) Find at least one eigenvalue–eigenvector pair of A .

(b) Rank-one matrix

Let $r \in \mathbb{R}^k$ with $k > 132$ and define $M = rr^T$.

- (i) How many nonzero eigenvalues does M have? Find them.
- (ii) Compute

$$\sum_{i=1}^k |\lambda_i|.$$